

Generative AI for Discovering Rare Astronomical Phenomena

Potential Faculty Advisors: Mr Klyne Smith (Not confirmed yet), Still waiting for responses

Research Question

How can Generative AI be utilized to synthesize realistic examples of rare astronomical phenomena to improve detection and classification in large-scale astrophysical surveys?

Summary

This research aims to leverage Generative AI to address the challenge of identifying rare astronomical phenomena, such as gravitational lensing, kilonovae, or fast radio bursts. These events are underrepresented in existing datasets due to their low occurrence rates, making it difficult for machine learning models to perform robust detection and classification. By developing models that generate realistic multimodal data—including images, spectra, and time-series—we aim to expand training datasets, improve detection pipelines, and enhance the understanding of these phenomena.

The project focuses on creating synthetic examples of rare events using a Generative Adversarial Network (GAN) or diffusion-based architecture trained on multimodal subsets of the **Multimodal Universe** dataset. The generated data will be validated for scientific accuracy using astrophysical metrics and integrated into ML workflows for tasks like transient detection and redshift prediction. This research aligns with the growing interest in leveraging AI for advancing space exploration and astrophysics.

Methodology

1. Data Collection and Preparation:

- Extract data on known rare events from the **Multimodal Universe** dataset, including images, spectra, and time-series data.
- Preprocess data to ensure consistency in scale, resolution, and modality.

2. Model Development:

- Implement a conditional Generative Adversarial Network (cGAN) or diffusion model for multimodal data generation.
- Incorporate a physics-informed loss function to ensure that generated data adheres to astrophysical principles.

3. Training and Validation:

- Train the model using GPU resources to handle the computational demands of generative architectures.

- Validate the generated data by comparing it to real observations, evaluating fidelity with astrophysical metrics like signal-to-noise ratio and feature accuracy.
- 4. Integration and Evaluation:**
- Use the synthetic data to train ML models for detecting rare events in astronomical surveys.
 - Test the enhanced models on benchmark tasks from the Multimodal Universe, such as transient classification or redshift prediction.
- 5. Collaborative Refinement:**
- Work with domain experts and astrophysicists to refine the generative model and ensure its outputs align with scientific expectations.

Resources Needed

- **Computational Infrastructure:**
High-performance GPU clusters for training generative models and processing large datasets.
NVIDIA 3060 should do but aiming higher (would be better to get access to a lab computer)
- **Access to Data:**
Subsets of the **Multimodal Universe** dataset, including cross-matched multimodal observations.

Annotated Bibliography

1. **Paper Title:** "The Multimodal Universe: Enabling Large-Scale Machine Learning with 100 TB of Astronomical Scientific Data"
 - **Source:** NeurIPS 2024 Datasets and Benchmarks Track
 - **Description:** Introduces the Multimodal Universe dataset, detailing its multimodal components (e.g., images, spectra, time-series) and machine learning benchmarks.
 - **Evaluation:** The dataset provides a rich source for training and validating generative models for astronomical data synthesis.
2. **Paper Title:** "Self-Supervised Learning for Multimodal Astrophysical Data"
 - **Source:** IEEE Transactions on Pattern Analysis and Machine Intelligence
 - **Description:** Discusses the application of contrastive learning techniques to align multimodal astrophysical datasets, forming the foundation for generative approaches.
 - **Evaluation:** Insights into multimodal pre training methods will inform the generative model architecture.
3. **Paper Title:** "Physics-Informed Generative Models for Scientific Data"
 - **Source:** Journal of Machine Learning Research

- **Description:** Explores how domain-specific physics can be embedded into generative AI frameworks, ensuring scientific plausibility.
- **Evaluation:** Provides a methodology for incorporating astrophysical constraints into generative AI models.